Euclidean Geometry In Mathematical Olympiads

Problem 1.37 (BAMO 1999/2)

277546
1607 posts

Mar 8, 2018, 9:23 PM • 3 #1

Problem 1.37 (BAMO 1999/2)
Let C be a circle in the xy-plane with center on the y-axis and passing through A = (0, a) and B = (0, b) with 0 < a < b. Let P be any other point on the circle, let Q be the intersection of the line through P and A with the x-axis, and let O = (0, 0). Prove that ∠BQP = ∠BOP

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Coordinate Solution

The equation for the circle Γ is $x^2+(y-\frac{a+b}{2})^2=(\frac{b-a}{2})^2$. Observe that $\angle BPQ=90$ by Thale's theorem. It follows that the circumcenter w of $\triangle BPQ$ lies on some point (k, b/2). Through some algebra, we realize that $k=\frac{\sqrt{b^2-2ab}}{2}$. Hence, $w(\frac{\sqrt{b^2-2ab}}{2},\frac{b}{2})$.Through the distance formula, observe that the radius of w is $\sqrt{\frac{b^2}{2}-\frac{ab}{2}}=r$. We now show that $wO=r$. This can be done by the distance formula, which indeed shows that $wO=\sqrt{\frac{b^2}{2}-\frac{ab}{2}}=r$. Thus, $\triangle BQP$ and $\triangle BOP$ share the same circumcircle, so $B,Q,O,$ and P are concyclic. The result follows. ■

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Mar 8, 2018, 9:34 PM • 3 #3

Man ■ is awesome.

Drunken_Master
327 posts

Mar 9, 2018, 3:05 AM • 9 #4

$\angle BOQ=\angle BPQ=90^\circ$, so $BPOQ$ is cyclic. Hence, done. □

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Mar 9, 2018, 6:40 AM • 5 #5

(Facepalm) why didn't I realize that.

Kuroshio
72 posts

Mar 24, 2020, 3:18 AM #6

First note that AB is the diameter of the circle.

$\angle BPQ=\angle BPA=90^\circ=\angle BOQ$
So points B,P,O,Q are con cyclic. The conclusion follows.

This post has been edited 2 times. Last edited by Kuroshio, Jul 14, 2020, 2:19 AM